

| Field/Parameter                                                                                 | Type             | Details                                                                                                                      | Example                                                                                                                                    | Mandatory           | Exception           |
|-------------------------------------------------------------------------------------------------|------------------|------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|---------------------|---------------------|
| Trace Ping Fields                                                                               |                  |                                                                                                                              |                                                                                                                                            |                     |                     |
| These fields represent the structure of each ping in the input trace.                           |                  |                                                                                                                              |                                                                                                                                            |                     |                     |
| latitude                                                                                        | int, float, None | Latitude of the ping in decimal degrees format. Must be within range [-90, 90], or None.                                     | 27.562381                                                                                                                                  | Yes                 | ValidationException |
| longitude                                                                                       | int, float, None | Longitude of the ping in decimal degrees format. Must be within range [-180, 180], or None.                                  | 77.52531                                                                                                                                   | Yes                 | ValidationException |
| timestamp                                                                                       | int              | Timestamp for the ping as an integer. Example: Unix epoch timestamp in milliseconds.                                         | 1706774541094                                                                                                                              | Yes                 | ValidationException |
| Function Parameters                                                                             |                  |                                                                                                                              |                                                                                                                                            |                     |                     |
| These fields represent the structure of each parameter for calculate_trace_similarity function. |                  |                                                                                                                              |                                                                                                                                            |                     |                     |
| trace_1                                                                                         | list             | The first trace for similarity calculation. Each entry is a list of [latitude, longitude, timestamp].                        | [<br>[77.12252044155173,<br>28.54777297844582,<br>1733032561000],<br><br>[77.12312348835866,<br>28.548800324214767,<br>1733032562000]<br>] | Yes                 | ValidationException |
| trace_2                                                                                         | list             | The second trace for similarity calculation. Each entry is a list of [latitude, longitude, timestamp].                       | [<br>[77.12456714586767,<br>28.553920851309357,<br>1733032561000],<br><br>[77.12421993710012,<br>28.55289355549519,<br>1733032562000]<br>] | Yes                 | ValidationException |
| distance_threshold                                                                              | float            | Maximum allowable spatial distance (in meters) for two pings to be considered overlapping. Example: 50.0                     | 100                                                                                                                                        | Yes                 | ValidationException |
| time_threshold                                                                                  | int              | Maximum allowable time difference (in milliseconds) for two pings to be considered overlapping. Example: 300000 (5 minutes). | 10000                                                                                                                                      | Yes                 | ValidationException |
| plot_map                                                                                        | bool             | Flag to indicate whether to generate a map visualization of the traces and their overlapping pings.                          | TRUE                                                                                                                                       | No (Default: False) | ValidationException |